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Murad Hossain

Nationality: Bangladeshi **Date of birth:** 12/04/1997 **Place of birth:** BARISHAL, Bangladesh

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ABOUT ME

Results-driven Mechanical Engineer with over 6 years of progressive experience in process development, production management, and industrial automation within the home appliance and boiler manufacturing sectors. Currently serving as Senior Assistant Director of Process Development at Walton Hi-Tech Industries PLC. Proven track record in reducing cooling performance test time, implementing AGV systems for lean manufacturing, integrating real-time monitoring (Pirani gauge + CAN-BUS), and driving continuous improvement through Six Sigma methodologies. Strong background in SolidWorks, thermal systems, vacuum technology, and IoT-based solutions.

WORK EXPERIENCE

Walton Hi-Tech Industries PLC. – Gazipur, Bangladesh

Process engineer

[03/05/2021 – Current]

Cooling performance test optimization, Vacuum level maintaining in Refrigerator circuit, Gas Charging Technology, Refrigerant selection, Process optimization, Tube joining technology, Traceability, Handling process optimization

New Model Boiler Company Limited – Gazipur, Bangladesh

Executive Engineer

[01/11/2019 – 30/04/2021]

Designing and manufacturing process supervision, different kinds of NDT, like radiographic and ultrasonic testing of welding joints, Directing heat treatment processes, hydraulic testing of boilers in accordance with the Boiler Regulation (1951), and conducting training to marketing officers, enhancing their technical knowledge of boiler engineering

Bangladesh Industrial Technical Assistance Centre (BITAC) – Dhaka, Bangladesh

Internship

[17/05/2018 – 11/06/2018]

Completed hands-on industrial training across key manufacturing and technical divisions including Machine Tools Shop, CNC Shop, Heat Treatment, Foundry, Welding & Fabrication, and Tool & Technology Institute (TTI).

EDUCATION AND TRAINING

Bachelor of Science in Mechanical Engineering

Rajshahi University of Engineering & Technology [25/01/2015 – 21/10/2019]

City: Rajshahi | **Country:** Bangladesh

Mechanical Engineering Drawing, Engineering Mechanics, Computer Programming Language, Mechanics of Solids, Heat Transfer, Fluid Mechanics, Engineering Metallurgy, Automobile Engineering, RAC etc.

PUBLICATIONS

[2025]

A Greener Future: IoT-Powered Smart Irrigation for Bangladesh This research presents an IoT-based automated irrigation system designed to optimize water usage and enhance crop productivity in regions facing water scarcity and unpredictable rainfall.

Authors: Md. Foysal, Mehedi Hasan, Murad Hossain, Most Salma Khatun | **Journal Name**: 1st International Conference on Agricultural Machinery and Bioresource Engineering (ICAMBE) 2025 | **Publisher**: Bangladesh Agriculture University

[2025]

Dynamics of Municipal Solid Waste in Urban Bangladesh: Compositional Shifts, GHG Emissions, and Policy Implications (2005–2041) We calculate trash generation will increase by 324% by 2041 with current GDP growth, found Chittagong and Dhaka as key hotspots. Our results offer a solid scientific foundation for policymakers to prioritize low-carbon waste management pathways in rapidly urbanizing economies.

Authors: Md. Foysal, Most Salma Khatun, Mehedi Hasan, Murad Hossain, Zannatul Fhardous Umama, Md. Fahad Hossain. | **Journal Name**: 4th National Research Project Contest 2025 | **Publisher**: Bangladesh University of Professionals

[2025]

Leveraging Climate-Smart Agriculture for Mitigation: Sustainable Solutions from Bangladesh In Bangladesh, shifting rainfall patterns and unpredictable seasons are already affecting farmers' ability to plan, plant, and harvest—putting food security at risk. The World Bank has identified climate-smart agriculture (CSA) as the key solution to safeguarding the future of farming. By focusing on sustainable practices that integrate climate adaptation and mitigation, CSA helps farmers improve their productivity while building resilience against climate shocks.

Authors: Md. Foysal, Most Salma Khatun, Mehedi Hasan, Murad Hossain, Zannatul Fhardous Umama | **Journal Name**: IAAS Asia Pacific Scientific Event 2025 | **Publisher**: IAAS Asia Pacific Scientific Event 2025

[2026]

FEA-Driven Topology Optimization for Lightweight Structural Design: A Computational Framework with Implicit Geometry Reconstruction Developed a topology-optimized Zamak 3 refrigerator hinge bracket, reducing mass **45.7%** while maintaining structural adequacy through FEA and DfM reconstruction.

Authors: Afsar Mahmud, Hridoy Bhadra, Murad Hossain | **Journal Name**: International Conference on Mechanical, Industrial and Materials Engineering | **Publisher**: International Conference on Mechanical, Industrial and Materials Engineering

[2019]

Simulation Of Modified Flipped Airfoil And Analysis Of Lift And Drag Coefficient Simulation and design of **NACA BE50** with different-sized dimple surfaces at different orientations, and analyzing the air flow profile at different angles of attack to get the best orientation to reduce the drag force

Authors: Murad Hossain, Prashanta Saha | **Journal Name**: Thesis Paper | **Publisher**: Rajshahi University of Engineering & Technology

SKILLS

Microsoft Office / 2D and 3D design software (Solidworks, AutoCad) / Ansys: Engineering Simulation / CSS & HTML / Programming: C Programming / Python (computer programming)

LANGUAGE SKILLS

Mother tongue(s): Bengali

Other language(s):

English

LISTENING C2 **READING** C1 **WRITING** B2

SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

CONFERENCES AND SEMINARS

[01/06/2025 – 30/05/2026]

Automotive Product Design Fundamentals of Plastic Trims and Body-in-White (BIW) components. The role and significance of reverse engineering in modern automotive design. Understanding industry-specific expectations, and the importance of professional networking

[14/12/2025 – 19/12/2025] Bangladesh Institute of Management (BIM)

Climate Change and Climate Finance Training, PKSF Training program on **climate change, Bangladesh's NAP and NDCs, Green Climate Fund (GCF) project development, results frameworks, climate finance, economic analysis, procurement, environmental and social safeguards, gender action plans, and GCF investment criteria**, including a field visit to the **CCDB Climate Center, Gazipur**.

CERTIFICATIONS

[VirtualTester Online, Tangix Design & Development AB, 25/12/2024]
3DSwymer Associate

[VirtualTester Online, Tangix Design & Development AB, 29/11/2024]
SOLIDWORKS Sustainability Associate

[VirtualTester Online, Tangix Design & Development AB, 28/11/2024]
SOLIDWORKS Additive Manufacturing Associate

[VirtualTester Online, Tangix Design & Development AB, 25/11/2024]
SOLIDWORKS Design Associate (CSWA)

[6sigmastudy-The global certification body for six sigma certification, 14/11/2024]
Six Sigma Yellow Belt

PROJECTS

[01/07/2025 – Current]
Cooling Circuits Temperature Profile Base Decision Making Reduction of power consumption approx. **158.4 MW** annually by reducing aging time till 15 minute

[14/08/2025 – 31/12/2025]
Oil Pulse Pneumatic Gun Implementation Improving operator ergonomics and increasing loosening torque from approximately 7 N·m to 10 N·m (43% improvement)

[06/08/2024 – 29/05/2025]
Thermal Image-Based Aging Test Model Developing Resulting **117.75 MW** energy save per annum

[27/01/2024 – 28/06/2025]
Pirani Gauge with the CAN HS System Implementation reduced chocking problem from **.13%(2021)** to **.02% (2025)**

[08/08/2023 – 20/02/2024]
60-Feet AGV Pilot Project Payload Analysis and Designing Lightweight Boxes

[20/10/2021 – 19/07/2023]
IR-Based Industrial Thermal Gun Temperature Measurement Model Implementation resulting **69.71 MW** reduction in power consumption annually

[19/07/2021 – 23/12/2021]
PET Belt Joining Strength Elevation Rather Changing Material Consumption Developed a thinner PET belt, increasing thermal-joint strength by **43%** without increasing material consumption.

[17/12/2019 – 23/11/2020]
3000 kg Shell Type Boiler Manufacturing Solving problem by designing and cutting optimum plate boiler grade plate dimension to avoid extra longitudinal seam

[17/08/2018 – 21/08/2019]
Simulation Of Modified Flipped Airfoil And Analysis Of Lift And Drag Coefficient Simulation and design of **NACA BE50** with different-sized dimple surfaces at different orientations, and analyzing the air flow profile at different angles of attack to get the best orientation to reduce the drag force

VOLUNTEERING

[08/09/2015 – 31/12/2015] Rajshahi
Active Citizen Affiliated with British Council Batch 43 to solve social issues through group efforts

[01/06/2010 – 30/11/2026] Barishal
Youth leader of the Red Crescent Society of Oxford Mission High School Served as first aid provider at the school and contributed to local community volunteer initiatives.

[23/11/2018 – 28/08/2019]
Vice President of Barishal Divisional Association Introducing rental base library for the needy students, arranged different occasions

HONOURS AND AWARDS

[30/10/2025] Walton Hi-Tech Industries PLC.
Best Employee of the Quarter Vacuum line modification , aging reduction project

[19/12/2022] Walton Hi-Tech Industries PLC.
Best Trainer Award welder skills development through training and minimizing joining leak percentage by 50%

[25/01/2015] Rajshahi University of Engineering & Technology
Merit Technical Government Scholarships

[18/01/2014] Notre Dame College
Perfect Attendance Award

[15/08/2011] Barisal Debating Society
Certification of Participation

[16/12/2011] Oxford Mission High School
1st Prize at Annual Sports and Cultural Competition (High Jump)

RECOMMENDATIONS

Name: **Dr. Md. Emdadul Hoque** | Thesis Supervisor and Professor
Simulation Of Modified Flipped Airfoil And Analysis Of Lift And Drag Coefficient
Email: mehoque@me.ruet.ac.bd | **Phone number:** (+880) 1713228615

Name: **Dr. Syed Mamun R Rasid** | Professor
Taught Engineering Mechanics-III and Fluid Mechanics-II
Email: mamun@me.ruet.ac.bd | **Phone number:** (+880) 1330546246